



30-Day AI Engineer Roadmap

Goal: Transition from Data Professional to AI Engineer

Time Commitment: 2 hours daily | **Start Date:** June 6, 2024

Week 1: AI Fundamentals & Deep Learning Basics

Milestone: Understand AI/ML pipeline & build first neural network

Days 1-3: AI Engineering Overview

Daily Tasks (40 min each):

- Watch: "AI Engineering vs Data Science" (YouTube)
- Read: AI system architecture basics
- Practice: Review Python OOP concepts

Resources: • Fast.ai Practical Deep Learning Course (Part 1) • "Hands-On Machine Learning" by Aurélien Géron (Ch. 10-11) • Google's Machine Learning Crash Course

Days 4-7: Neural Networks & TensorFlow

Daily Tasks:

- 30 min: TensorFlow basics tutorials
- 60 min: Code along neural network examples
- 30 min: Document learnings in Jupyter notebooks

Weekend Project: Build a digit classifier using MNIST dataset

Week 2: Deep Learning Frameworks & Computer Vision

Milestone: Deploy first computer vision model

Days 8-11: PyTorch & Computer Vision

Daily Tasks:

- 45 min: PyTorch fundamentals
- 45 min: CNN architecture study
- 30 min: Image preprocessing techniques

Days 12-14: Model Training & Optimization

Daily Tasks:

- 60 min: Transfer learning implementation
- 30 min: Hyperparameter tuning
- 30 min: Model evaluation metrics

Weekend Project: Create an image classification API using FastAPI

Resources: • PyTorch tutorials official website • CS231n Stanford lectures (selected videos) • Kaggle Learn Computer Vision

Week 3: NLP & Large Language Models

 **Milestone: Build a text generation application**

Days 15-18: NLP Fundamentals & Transformers

Daily Tasks:

- 45 min: NLP preprocessing (tokenization, embeddings)
- 45 min: Transformer architecture study
- 30 min: Hugging Face tutorials

Days 19-21: Working with Pre-trained Models

Daily Tasks:

- 60 min: Fine-tuning BERT/GPT models
- 30 min: Prompt engineering basics
- 30 min: LangChain introduction

Weekend Project: Build a chatbot using OpenAI API or open-source LLM

Resources: • Hugging Face NLP Course • "Natural Language Processing with Transformers" book • Andrew Ng's ChatGPT Prompt Engineering course

Week 4: MLOps & Production Deployment

 **Milestone: Deploy end-to-end AI application**

Days 22-25: MLOps Essentials

Daily Tasks:

- 45 min: Docker containerization for ML
- 45 min: Model versioning with MLflow
- 30 min: CI/CD for ML pipelines

Days 26-28: Cloud Deployment & Monitoring

Daily Tasks:

- 60 min: Deploy models on AWS/GCP/Azure
- 30 min: Model monitoring setup
- 30 min: Performance optimization

Days 29-30: Integration & Portfolio

Final Project: End-to-end AI application with:

- Data pipeline (using existing SQL/Python skills)
- Model training & evaluation
- API deployment
- Simple web interface
- Monitoring dashboard

Resources: • "Building Machine Learning Powered Applications" by Anouar Malainine • MLOps Specialization on Coursera • Made With ML MLOps course

Final Outcome

You will have:

-  Built 5+ AI projects (CV, NLP, deployment)

- ☒ Hands-on experience with TensorFlow, PyTorch, Transformers
- ☒ Deployed models to production
- ☒ Portfolio showcasing end-to-end AI solutions
- ☒ Strong foundation for AI engineering roles

Next Steps:

- Contribute to open-source AI projects
- Specialize in either Computer Vision or NLP
- Learn advanced topics: Reinforcement Learning, GANs
- Build domain-specific AI solutions

Remember: Consistency beats intensity. 2 hours daily for 30 days = 60 hours of focused learning!